USN | | | | | | | | | | [23CSBSMA11]

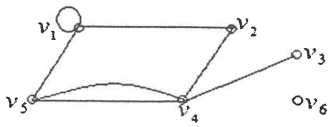
FIRST SEMESTER B.Tech EXAMINATION, JANUARY – 2024

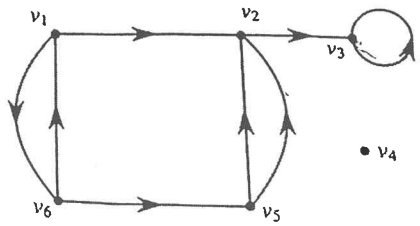
DISCRETE MATHEMATICS FOR COMPUTER SCIENCE

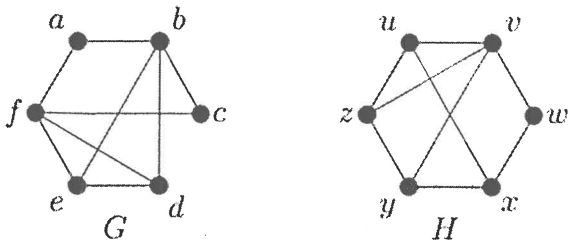
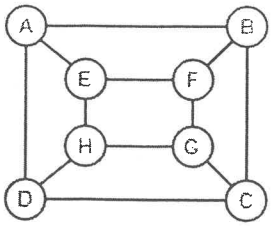
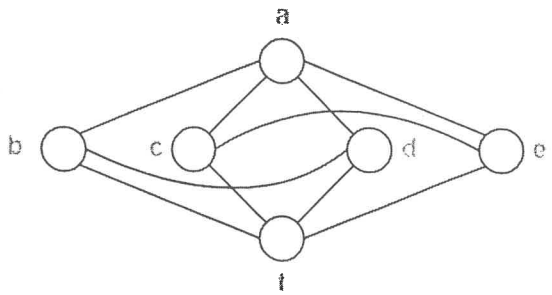
Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A	**05 X 02=10**		
1.	**a.**	Define Disjunction and write its truth table.	L-1	CO1	**02**
	b.	Determine the sequence generated by the function $(3+x)^3$.	L-2	CO2	**02**
	c.	In a class, there are 4 boys and 10 girls if a boy and a girl have to be chosen for the class monitor, in how many ways the students can choose class monitor?	L-2	CO3	**02**
	d.	Compute the degree of each vertex of the following graph: 	L-2	CO6	**02**
	e.	Define NOR logic gate.	L-1	CO5	**02**
		PART-B	**06 X 05=30**		
2.	**a.**	Prove the logical equivalence $(\neg p \wedge q) \vee \neg (p \vee q) \Leftrightarrow \neg p$.	L-3	CO1	**05**
		OR			
	b.	Apply rules of inference to show that r is logically derived from $p \rightarrow q$, $q \rightarrow r$, and p.	L-3	CO1	**05**
3.	**a.**	Give the direct proof for the following statement: *"The sum of two even numbers is even"*	L-3	CO2	**05**
		OR			
	b.	Show that if seven colors are used to paint 50 bicycles, at least eight bicycles will be the same color.	L-3	CO2	**05**
4.	**a.**	The number of virus-affected files in a system is 1000 [to start with] and it increases 250% every two hours. Use recurrence relation to determine the number of virus affected files in the system after one day.	L-3	CO3	**05**
		OR			
	b.	Determine the coefficient of x^{27} in $(x^4+x^3+x^5+\cdots)^5$	L-3	CO3	**05**
5.	**a.**	Let $A=\{2,3,4\}$ and $B=\{4,5\}$. Compute **i.** $A \times B$ **ii.** $B \times A$ **iii.** $\lvert A \times B \rvert$ **iv.** $\lvert B \times A \rvert$	L-2	CO4	**05**
		OR			
	b.	Show that the union of two subgroups of a group G need not be a subgroup.	L-2	CO4	**05**

6.	**a.**	Sketch the K-maps for the following Boolean expressions **i.** $xy + \bar{x}y$ **ii.** $x\bar{y} + \bar{x}y$ **iii.** $x\bar{y} + \bar{x}y + \bar{x}\,\bar{y}$	L-3	CO5	**05**
		OR			
	b.	Express the Boolean function $F = x + yz$ as a sum of minterms.	L-3	CO5	**05**
7.	**a.**	Is it possible to construct a graph with 12 vertices such that 2 of the vertices have degree 3 and remaining vertices are having degree 4?	L-3	CO6	**05**
		OR			
	b.	Define In-degree and Out-degree of a vertex. Compute In-degree and Out-degree of each vertex for the following directed graph:  	L-2	CO6	**05**
		PART-C	**06 X 10=60**		
8.	**a.**	Using truth table, discuss the nature of the following compound proposition: $\{(p \vee q) \wedge [(p \rightarrow r) \wedge (q \rightarrow r)]\} \rightarrow r$	L-3	CO1	**10**
		OR			
	b.	Obtain the Principal Disjunctive normal form (PDNF) and Principal conjunctive normal form (PCNF) of $P \vee (\neg P \rightarrow (Q \vee (\neg Q \rightarrow R)))$.	L-3	CO1	**10**
9.	**a.**	Solve the recurrence relation $F_{n+2} = F_{n+1} + F_n \; \forall n \geq 0$ given $F_0 = 0,\ F_1 = 1$	L-3	CO2	**10**
		OR			
	b.	Using principle of mathematical induction, prove that $1^3 + 2^3 + 3^3 + \cdots + n^3 = \frac{n^2(n+1)^2}{4}, \quad \forall\, n \geq 1$	L-3	CO2	**10**
10.	**a.**	Determine the generating functions for the following sequences: **i.** 1, 2, 3, 4,... **ii.** 1, -2, 3, -4,... **iii.** $1^2, 2^2, 3^2, 4^2$,...	L-3	CO3	**10**
		OR			
	b.	**i.** How many passwords of length 4 having four upper case letters are generated such that no letters can be duplicated in a password? **ii.** If 13 people are assembled in a room, show that at least two of them must have their birthdays in the same month.	L-3	CO3	**10**

11.	**a.**	Show that, the set of non-zero real numbers is an abelian group with respect to multiplication operation.	L-3	CO4	**10**
		OR			
	b.	Show that the set of all integers is a group with respect to addition (+) operation.	L-3	CO4	**10**
12.	**a.**	Verify the validity of the following distributive laws through truth table: **i.** $x(y+z) = xy + xz$ **ii.** $x + (yz) = (x+y) + (x+z)$	L-3	CO5	**10**
		OR			
	b.	Sketch the logic circuits that produce the following outputs: **i.** $(xy + xz + yz)$ **ii.** $(x + y + z)\,(\bar{x}\,\bar{y}\,\bar{z}\,)$	L-3	CO5	**10**
13.	**a.**	**i.** Verify whether below two graphs G and H are Isomorphic or not?  **ii.** A connected planar graph G has 9 vertices with degrees 2, 2, 3, 3, 3, 4, 5, 6, 6. Compute the number of regions of G.	L-3	CO6	**10**
		OR			
	b.	**i.** Determine whether the following graph contains a Hamiltonian path and Hamiltonian Cycle?  **ii.** Define Chromatic number and hence determine the Chromatic number for the following Graph:  	L-3	CO6	**10**